
Project Report: Smartphone Price Prediction

Prepared for:

Dr. Fatma ElShehaby

Data Science and Methodology
(CSE221)

Prepared by:

Abdelrahman El-Khashab 23-101182

Abdullah Helmy 23-101239

Mariam Hassan 23-101309

Maryam Rageh 23-101297

Mohamed Elshabouri 23-101334

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1. Abstract

2. Introduction

Background and Context:

Literature Review:

Smartphone price prediction has been an active area of applied machine learning research, framing the problem as both regression and classification — the same dual formulation adopted in this project.

Studies have compared a wide range of algorithms including Decision Tree, Random Forest, SVM, KNN, and Naive Bayes for predicting mobile price ranges, with tree-based ensembles consistently outperforming linear classifiers (ICMLT). More recent work has explored neural networks, with an MLP model achieving 92% accuracy in fair price prediction (Rezayan et al.), outperforming SVM and Logistic Regression baselines. This suggests deep learning represents a meaningful ceiling above tree-based methods and is a natural next step for this project.

A consistent limitation across the literature is the same one encountered here: hardware specifications alone do not fully explain price, and the absence of brand as a feature constrains model performance regardless of algorithm choice.

Project Objectives:

Clearly state the specific goals and objectives of your project.

3. Materials and Methods

3.1 Data Description

3.2 Data Cleaning

3.3 Methodology

3.4 Model Evaluation Metrics

4. Results

5. Conclusions

5.1 Summary of findings

5.1.1 Regression

5.1.2 Classification

Devices were assigned to price tiers: **Budget** (<₹8k), **Mid-Range** (₹8k–25k), **Premium** (>₹25k).

5.2 Recommendations

5.3 Limitations

5.4 Future Work

6. Acknowledgements

Team Contributions

Acknowledge the contributions of each team member to the project.

7. References